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Homework 7

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1. NEWMAN EXERCISE 8.15 A,B

We look to model the total energy in a double pendulum system. The motion of the double pendulum is mathematically dependent on the two angles θ_1 and θ_2 as well as the height of the masses measured from the level of the pivot, h_1 and h_2. The heights can be given by

$$h_1 = -l\cos\theta_1 \tag{1}$$

$$h_2 = -l(\cos(\theta_1) + \cos(\theta_2)) \tag{2}$$

where l is the length of the pendulum. To effectively model the system, we will construct the Lagrangian, which after plugging into the Euler-Lagrange equation, will spit out two second order differential equations. From these two, we can create four first order equations that RK4 can solve.

The potential energy of the system is given by

$$V = -mgl(2\cos\theta_1 + cos\theta_2) \tag{3}$$

and the total kinetic energy is given by

$$T = ml^2[\dot{\theta_1}^2 + \frac{1}{2}\dot{\theta_2}^2 + \dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2)] \tag{4}$$

From this we can construct the Lagrangian

$$\mathcal{L} = T - V, \tag{5}$$

$$\mathcal{L} = ml^2[\dot{\theta_1}^2 + \frac{1}{2}\dot{\theta_2}^2 + \dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2)] + mgl(2\cos\theta_1 + cos\theta_2) \tag{6}$$

and after plugging this into the Euler-Lagrange equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial\mathcal{L}}{\partial\dot{\theta_1}} = \frac{\partial\mathcal{L}}{\partial\theta_1}, \tag{7}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial\mathcal{L}}{\partial\dot{\theta_2}} = \frac{\partial\mathcal{L}}{\partial\theta_2} \tag{8}$$

we output the second order equations

$$2\ddot{\theta_1} + \ddot{\theta_2}\cos\theta_1 - \theta_2 + \dot{\theta_2}^2\sin\theta_1 - \theta_2 + 2\frac{g}{l}\sin\theta_1 = 0, \tag{9}$$

$$\ddot{\theta_2} + \ddot{\theta_1}\cos\theta_1 - \theta_2 - \dot{\theta_1}^2\sin(\theta_1 - \theta_2) + \frac{g}{l}\sin\theta_2 = 0. \tag{10}$$

From here, we can create four first order equations

$$\dot{\theta_1} = \omega_1 \tag{11}$$

$$\dot{\theta_2} = \omega_2 \tag{12}$$

$$\dot{\omega_1} = \frac{-\omega_1^2\sin 2\theta_1 - 2\theta_2 + 2\omega_2^2\sin\theta_1 - \theta_2 + \frac{g}{l}[\sin\theta_1 - 2\theta_2 + 3\sin\theta_1]}{3 - \cos(2\theta_1 - 2\theta_2)} \tag{13}$$

$$\dot{\omega_2} = \frac{4\omega_1^2\sin\theta_1 - 2\theta_2 + \omega_2^2\sin 2\theta_1 - 2\theta_2 + \frac{g}{l}[\sin 2\theta_1 - \theta_2 - \sin\theta_2]}{3 - \cos(2\theta_1 - 2\theta_2)} \tag{14}$$

The total energy for the system can be calculated from

$$E = T + V \tag{15}$$

$$E = ml^2[\dot{\theta_1}^2 + \frac{1}{2}\dot{\theta_2}^2 + \dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2)] - mgl(2\cos\theta_1 + cos\theta_2). \tag{16}$$

We can now create a python program to solve these four equations and plot the total energy over time. We begin with pseudo code

```
import dependencies

initialize constants

# define the function that we wish to solve
def f(r,t):
    # since we are solving a system of coupled
    # ode's, we will take in a vector r that
    # contains all four functions

define interval
define stepsize
calculate height

initial conditions

Runge-Kutta 4 algorithm
```

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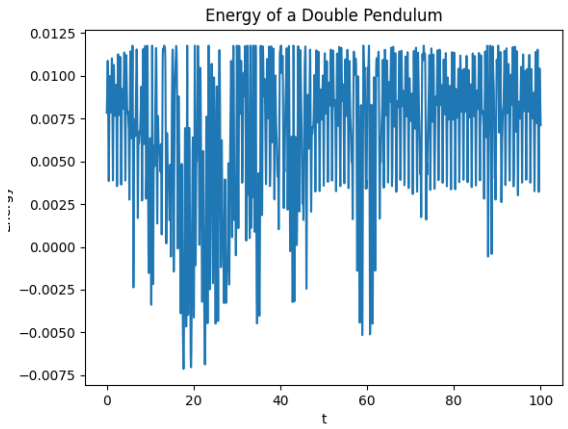
```
    return values of theta and omega

calculuate total energy
using theta and omega values

plot energy vs. time
```

For our initial conditions, $\omega_1 = \omega_2 = 0$ and $\theta_1 = \theta_2 = \frac{\pi}{2}$ and we will be looking at from $t = 0$ to $t = 100$. Figure 1 is our outputted plot for the total energy.



should be smaller

FIG. 1: Total energy for a double pendulum system.

The total energy in the system is shown to vary by roughly 10^-3J. With these initial conditions, we expect the total energy to be constant at 0, however, due to computer imprecision, we see noise and variation. The number of steps in this output is $N = 1,000,000$. Increasing the number of steps still outputs the varying pattern with strong spikes in certain parts of the graph. Perhaps with a stronger computer, the number of steps can be increased even further and the plot will converge to near constant 0.

2. NEWMAN EXERCISE 8.14

We look to model quantum oscillators using the time-independent, one-dimensional Schrodinger equation. The equation is given by

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} + V(x)\psi(x) = E\psi(x) \tag{17}$$

where $\psi(x)$ is the wavefunction, V(x) is the potential energy at position x, and E is the total energy of the particle. Turning this linear, 2nd order ode into a system of two coupled first order ode's,

$$\frac{\mathrm{d}\psi}{\mathrm{d}x} = \phi, \tag{18}$$

$$\frac{\mathrm{d}\theta}{\mathrm{d}x} = \frac{2m}{\hbar^2}[V(x) - E]\psi(x). \tag{19}$$

We will be looking at two cases, a harmonic oscillator and an anharmonic oscillator. The variation comes in their defined potential functions. For a harmonic oscillator,

$$V(x) = \frac{V_0 x^2}{a^2} \tag{20}$$

while for an aharmonic oscillator

$$V(x) = \frac{V_0 x^4}{a^4} \tag{21}$$

where a is a constant.

For a quantum particle that behaves like a harmonic oscillator (or anharmonic), we can look at the system like a boundary value problem. The wavefunction for the particle, which predicts a probability that the particle will be in a given position, must fit within the boundary of the particle's potential curve. Mathematically, we find that it is the variation in a particle's total energy, E, that determines if ψ is a valid solution to the boundary value problem. In other words, there are discrete values for E, or energy states, where the wavefunction at the boundary is equal to 0. This lends to the idea that energy is in fact quantized and comes in discrete levels. We will look for the value of these energy levels (specifically $n = 0$, also called the ground state, $n = 1$, and $n = 2$) for both the harmonic and anharmonic oscillator. We will also look to plot the wavefunctions for these three states of the anharmonic oscillator. To do this, we will first need to normalize the wavefunctions. To do this, we will need to solve Equation 22 for some coefficient that ensures the inner product between the wavefunction and itself outputs a valid probability.

$$\int_{-\infty}^{\infty} |\psi(x)|^2 \, dx = 1 \tag{22}$$

For our initial conditions, $\psi = 0$ and $\phi = 1$. We begin with pseudo code:

```
import dependencies

initialize constants

#define the potential
def V(x):
    # potential function

# define the function that we wish to solve
def f(r,x,E):
    # this will be the wavefunction
```

```
define interval
define stepsize
calculate height

initial conditions

Runge-Kutta 4 algorithm
    Will solve the Schrodinger equation to
    output our wavefunction

# We will use the secant method to find
# roots of the wavefunction; this is where
# we begin varying the energy until
# the wavefunction is a valid
# solution to the boundary
E1 = 0
E2 = e
Find energy levels

define normalization method
    Will output a normalizing coefficient

multiply values of wavefunction by the
coefficient

plot wavefunctions vs. x
```

For the harmonic oscillator, the energy levels are found in Table I. For the anharmonic oscillator, the energy levels are found in Table II.

$$E_0 = 138.024eV$$
$$E_1 = 414.072eV$$
$$E_2 = 690.120eV$$

TABLE I: Energy levels for a quantum harmonic oscillator

$$E_0 = 205.307eV$$
$$E_1 = 735.691eV$$
$$E_2 = 1443.570eV$$

TABLE II: Energy levels for a quantum anharmonic oscillator

The plot of the three wavefunctions can be found in Figure 2.

Looking at Figure 2, the wave number is directly correlated to the energy state of the oscillator. We can see that as the energy state increases, so does the wave number. This again lends to the idea of energy being quantized. A wave that is contained within a boundary must have an integer wave number. This is only possible when we admit only specific energy values for the wavefunction. These energy values are the eigenvalues of the system. Energy is discrete and thus quantized.

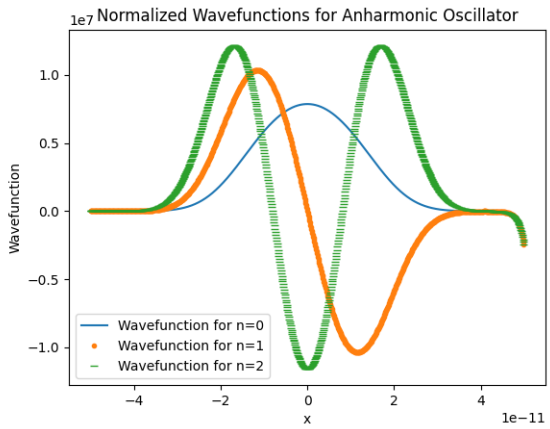


FIG. 2: Plot of three energy states of a wavefunction for a quantum anharmonic oscillator.

3. SURVEY

The problem set took roughly 10 hours. Exercise 8.14 was long but actually very enjoyable and rewarding. Exercise 8.15 seemed to give me quite a bit of issue when it came to the total energy converging no matter what value of N I chose. I believe it was due to either some sort of type error when inputting the extremely long equations, or there must be some extra numerical imprecision in my code that I'm not accounting for.

8.15

Computational Physics/Astrophysics, Winter 2023:

Grading Rubrics [1]

Haverford College, Prof. Daniel Grin

For coding assignments, roughly 25 points will be available per problem.

1. Does the program complete without crashing in a reasonable time frame? If yes, up to +3 points. 3

2. Does the program use the exact program files given (if given), and produce an answer in the specified format? If yes, +1 points 1

3. Does the code follow the problem specifications (i.e numerical method; output requested etc.) Up to +2 points 2

4. Is the answer correct? Up to+4points

   -1 error should be smaller

5. Is the code readable? Up to+2points 2

. 5.1. Are variables named reasonably?

. 5.2. Are the user-functions and imports used?

. 5.3. Are units explained (if necessary)?

. 5.4. Are algorithms found on the internet/book/etc. properly attributed?

[1] Inspired by rubric of D. Narayanan, U. Florida, and C. Cooksey, U. Hawaii

6. Is the code well documented? +3points 3

. 6.1. Is the code author named?

. 6.2. Are the functions described and ambiguous variables defined?

. 6.3. Is the code functionality (i.e. can I run it easily enough?) documented?

7. LaTeX writeup (up to 10 points) 10

- Are key figures and numbers from the problem given? (3 points)
- Is a brief explanation of physical context given? (2 points)
- If relevant, are helpful analytic scalings or known solutions given? (1 point)
- Are 3-4 key equations listed (preferably the ones solved in the programming assignment) and algorithms named? (2 points)
- Are collaborators clearly acknowledged? (1 point)
- Are any outside references appropriately cited? (1 point)

Note, even if (1), (2), (3), or (4) are not correct, one can still obtain many points via (5), (6), and (7).

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8.14

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Haverford College, Prof. Daniel Grin

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3. Does the code follow the problem specifications (i.e numerical method; output requested etc.) Up to +2 points 2

4. Is the answer correct? Up to+4points 4

5. Is the code readable? Up to+2points 2

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